

Week-03

Multimedia concepts

Scalable Vector Graphics (SVG)

Introduction - S

● Scalable

- Scalable Vector Graphics is an XML grammar for scalable graphics that can be used as an XML namespace.
- Scalable graphics allows for uniform dynamic pixel sizing for graphics.
- Graphic content can be stand-alone, referenced or included inside of other SVG graphics allowing for a complex illustration to be built in parts and possibly by several different people
- Scalable to allow for different display resolutions i.e. content magnification to aid people with low vision.

● Scalable Vector Graphics and the Web

- Scalable meaning the technology can grow to include a large number of users, files and/or application and still be efficient and effective.

Introduction - V

● Vector

- Vector graphics have geometric objects like lines and curves giving greater flexibility than raster-only formats (JPG, PNG) that store information for each and every pixel of the graphic.
- In general vector formats can include raster images as well as geometric objects and combine them with vector information.

Introduction - G

● Graphics

- Most existing XML grammars represent textual information or raw data and typically provide simply rudimentary graphical capabilities which are often less capable than the HTML 'img' element.
- SVG provides a rich, structured description of vector and mixed vector/raster graphics allowing it to be used in a stand-alone function or as an XML namespace with other grammars.

Introduction

● So what is SVG

- SVG is a web format that allows content developers to create two-dimensional graphics in a standard way by using an XML grammar.

● Why use SVG

- Size Matters
 - File Size
 - Resizing

● It's still XML

- Versatile
 - Static Graphics Rendering
 - Self-contained Applications
 - Server-based applications
- Intuitive

The SVG tag

First, we have to talk about the **svg tag** itself. This tag contains the image elements and defines the frame of our image. It sets the inner size and the outer size of the image.

The **width** and **height** property define how much space the image takes up in the browser. There's often a **viewBox** property as well. This defines a coordinate system for the elements inside the image. These two can be confusing because they both define a size.

Example (html and SVG)-Rectangle

```
<!DOCTYPE html>
<html>
  <head>
    <title>A page</title>
  </head>
  <body>
    <svg width="10" height="10">
      <rect x="0" y="0" width="10" height="10" fill="blue" />
    </svg>
  </body>
</html>
```

SVG example-1

```
<svg version="1.1"  
  width="300" height="200"  
  xmlns="http://www.w3.org/2000/svg">  
  
  <rect width="100%" height="100%" fill="red" />  
  
  <circle cx="150" cy="100" r="80" fill="green" />  
  
  <text x="150" y="125" font-size="60" text-anchor="middle" fill="white">SVG</text>  
  
</svg>
```



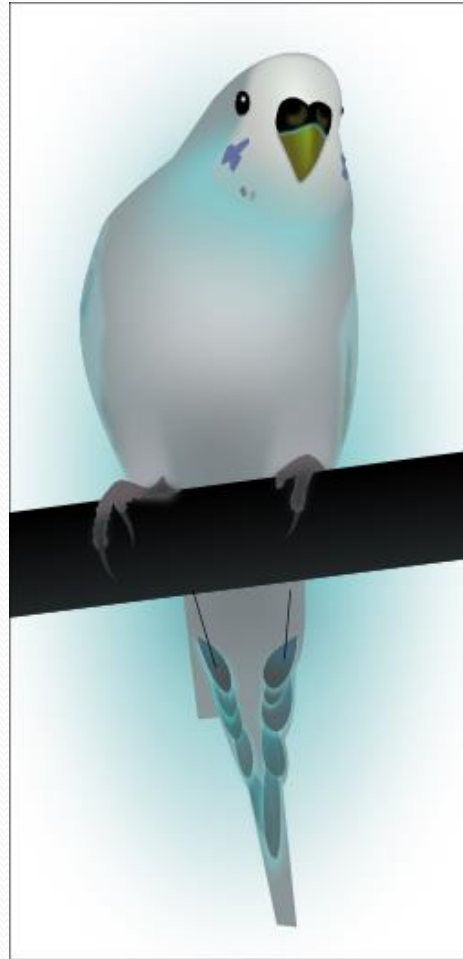
Example 2-Build a Tree with SVG

```
<html>
  <svg width="200" height="200" viewBox="-100 -100 200 200">
    <polygon points="0,0 80,120 -80,120" fill="#234236" />
    <polygon points="0,-40 60,60 -60,60" fill="#0C5C4C" />
    <polygon points="0,-80 40,0 -40,0" fill="#38755B" />
    <rect x="-20" y="120" width="40" height="30" fill="brown" />
  </svg>
</html>
```



<https://www.freecodecamp.org/news/svg-tutorial-learn-to-code-images/>

SVG example-3



<http://lds-jedi.deviantart.com/art/WinterSky-Vector-60966106>

Elements

Element—The element type name can be thought of as the tag name. It determines how the tag will function and consists of one empty (also known as “closed”) tag or two enclosing tags (a start tag and an end tag).• Attribute—“Attribute” describes an element by giving further infor

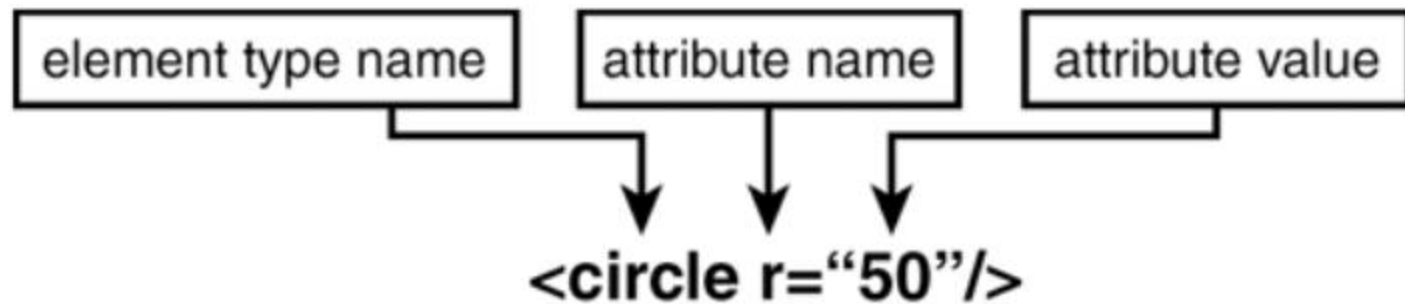


Figure 2.1. Diagram showing the relationship of the element, attribute name, and attribute value to each other within a sample SVG tag.

Common SVG Elements

The most used ones are

- `text`: creates a text element
- `circle`: creates a circle
- `rect`: creates a rectangle
- `line`: creates a line
- `path`: create a path between two points
- `textPath`: create a path between two points, and a linked text element
- `polygon`: allows to create any kind of polygon
- `g`: groups separate elements

Source: <https://flaviocopes.com/svg/>

text

- The text element adds text. The text can be selected using the mouse. x and y define the starting point of the text

```
<svg>  
  <text x="5" y="30">A nice rectangle</text>  
</svg>
```

A nice rectangle

SVG - Stroke

drawing line thickness and attributes

attributes	Stroke Type & Description
1	stroke – defines color of text, line or outline of any element.
2	stroke-width – defines thickness of text, line or outline of any element.
3	stroke-linecap – defines different types of ending of a line or outline of any path.
4	stroke-dasharray – used to create dashed lines

Sample SVG Stroke

Using stroke-width:



Using stroke-linecap:



Using stroke-dasharray:



circle

- Define a circle. `cx` and `cy` are the center coordinates, and `r` is the radius. `fill` is a common attribute and represents the figure color.

```
<svg>  
  <circle cx="50" cy="50" r="50" fill="#529fca" />  
</svg>
```



rect

- Defines a rectangle. x, y are the starting coordinates, with width and height.

```
<svg>  
  <rect x="0" y="0" width="100" height="100" fill="#529fca"  
/>  
</svg>
```



line

- x_1 and y_1 define the starting coordinates. x_2 and y_2 define the ending coordinates. `stroke` is a common attribute and represents the line color.

```
<svg>  
  <line x1="0" y1="0" x2="100" y2="100" stroke="#529fca" />  
</svg>
```



path

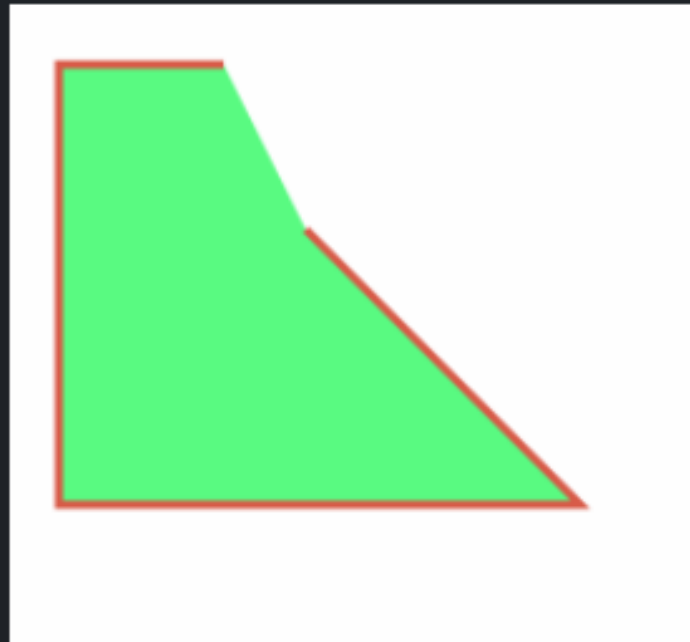
- A path is a sequence of lines and curves. It's the most powerful tool to draw using SVG, and as such it's the most complex.

`d` contains the directions commands. These commands start with the command name, and a set of coordinates:

- `M` means Move, it accepts a set of coordinates x, y
- `L` means Line, it accepts a set of coordinates x, y to draw the line to
- `H` is an Horizontal Line, it only accept an x coordinate
- `V` is a Vertical Line, it only accept an y coordinate
- `Z` means Close Path, puts a line back to the start
- `A` means Arch, it needs a whole tutorial on its own
- `Q` is a quadratic Bezier curve, again it needs a whole tutorial on its own

Path Example

```
<svg height="300" width="300">  
  <path d="M 100 100 L 200 200 H 10 V 40 H 70"  
        fill="#59fa81" stroke="#d85b49" stroke-width="3" />  
</svg>
```



textPath

- Adds a text along the shape of a path element.

```
<svg viewBox="0 0 1000 600"
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink">
  <defs>
    <path id="MyPath" d="M 20 40 Q 260 240 400 500" />
  </defs>
  <use xlink:href="#MyPath" fill="none" stroke="#59fa81" />
  <text font-family="Courier New" font-size="42.5">
    <textPath xlink:href="#MyPath">
      Wow such a nice SVG tut
    </textPath>
  </text>
</svg>
```

الكود للاطلاع فقط

Wow such a nice SVG tut

polygon

- Draw any random polygon with polygon. `points` represents a set of x, y coordinates the polygon should link:

```
<svg>  
  <polygon points="9.9, 1.1, 3.3, 21.78, 19.8, 8.58, 0, 8.58, 16.5, 21.78" />  
</svg>
```



g “group”

- Using the g element you can group multiple elements:

```
<svg width="200" height="200">
  <rect x="0" y="0" width="100" height="100" fill="#529fca" />
  <g id="my-group">
    <rect x="0" y="100" width="100" height="100" fill="#59fa81" />
    <rect x="100" y="0" width="100" height="100" fill="#ad4a3d" />
  </g>
</svg>
```



Some SVG transforms

Transform	Meaning
<code>translate(x, y)</code>	translate x, y units in respective directions.
<code>rotate(alpha, x, y)</code>	rotate alpha degrees around point (x,y).
<code>rotate(alpha)</code>	<code>rotate(alpha, o, o)</code> .
<code>scale(k)</code>	Scales the object with relation to the SVG origin with a factor k. This also scales the distance of the object to the origin.
<code>matrix(a,b,c,d,e,f)</code>	All transformations can be expressed as matrices, see Jenkins SVG tutorial.

More elements ...

● Check the tutorial at this page:

<https://flaviocopes.com/svg/>

Inkscape Tutorial

based on slides by: Simon Andrews

What is Inkscape?

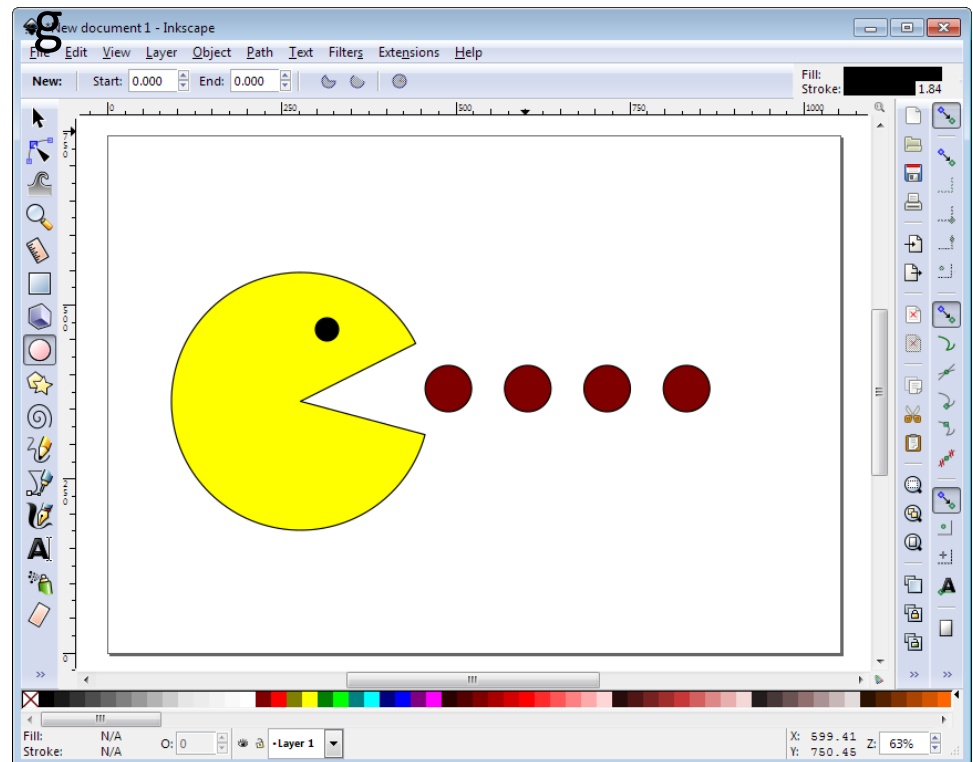
- Vector Graphics Editor
- Free Software
- Cross Platform
- Easy to use
- Good for:
 - Drawing
- Not for:
 - Bitmap editing



INKSCAPE

Draw Freely.

www.inkscape.org



Moving around

- Panning
 - Scroll bars on bottom / right
 - Scroll up/down, Shift+scroll for left/right
- Zooming in / out
 -  Click to zoom in, shift+click to zoom out
 - Control + Scroll Up/Down to zoom in/out to cursor
- Shortcuts
 -  Fit page,  drawing,  selection in window

The main toolbar



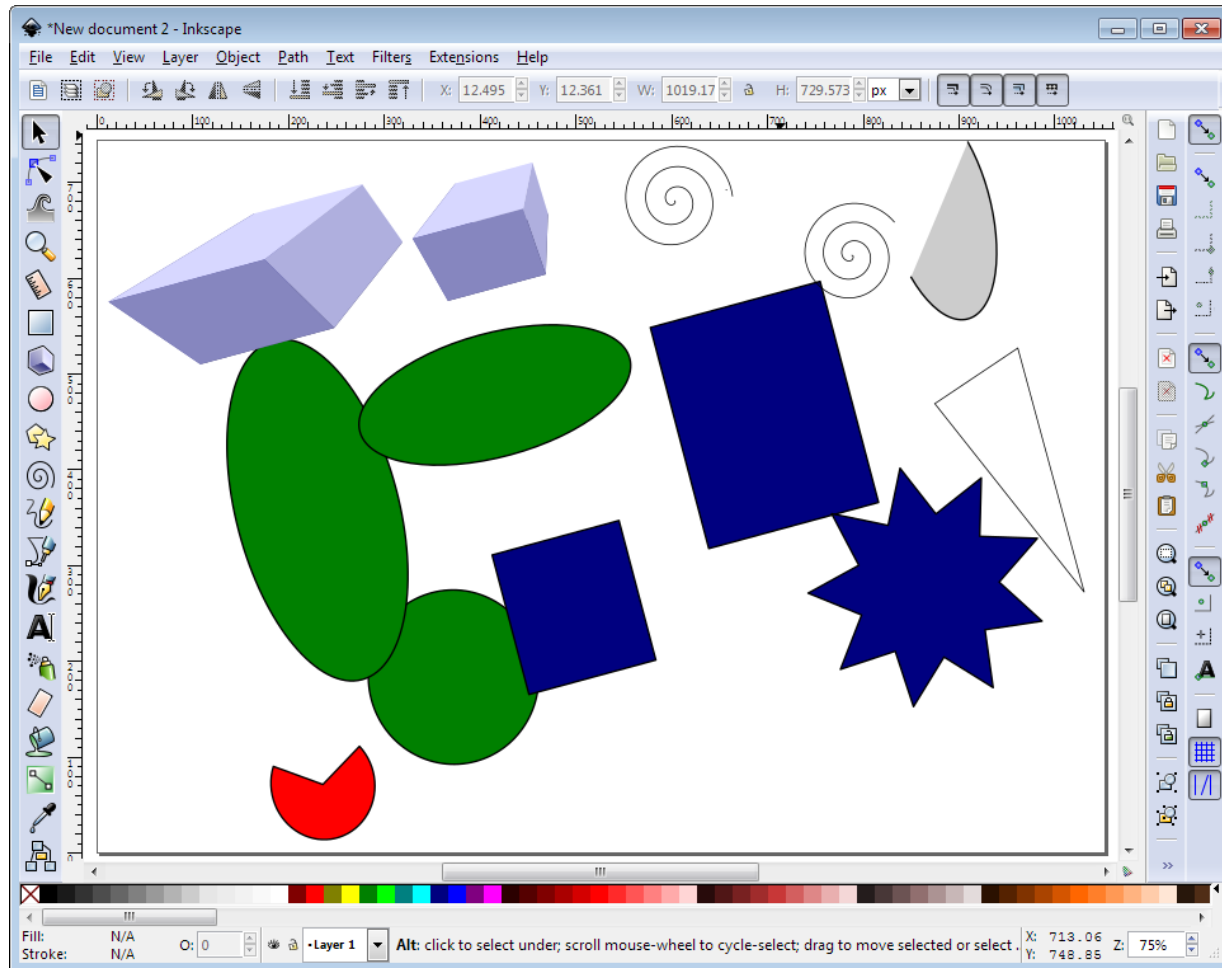
- Selection tool
- Edit nodes tool
- Sculpt tool
- Zoom tool
- Measurement tool
- Make rectangles
- Make 3D boxes
- Make ellipses / arcs
- Make polygons / stars
- Make spirals
- Draw freehand lines
- Draw straight lines / curves
- Calligraphy tool
- Add text
- Sculpt with spray
- Erase
- Fill
- Edit gradients

- Shortcuts
 - F1 Selection tool
 - F2 Edit nodes tool
 - F3 Zoom tool
 - F4 Rectangles
 - F5 Ellipses
 - F6 Freehand lines
 - F8 Text
 - F9 Spirals

Creating basic shapes

- Select tool from toolbar
- Click and drag on canvas
 - Box selects the bounds of the new shape
 - Colours are remembered from the last shape
- Basic options appear in top toolbar
 - Number of spokes on stars
 - Rounded corners on rectangles
 - Circle vs segment vs arc

Creating basic shapes



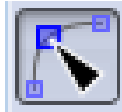
Creating basic shapes

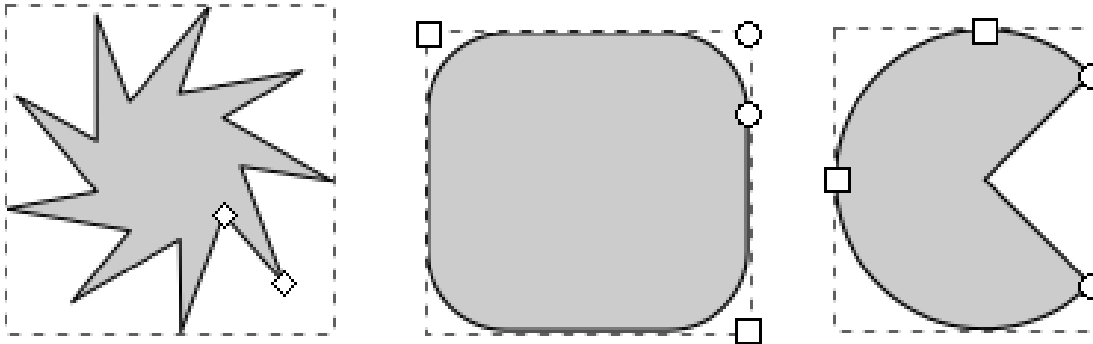
- Modifiers – press keys whilst drawing
 - **Control** = Constrain height/width ratio
(Easiest way to make **circles** / **squares**)
 - **Shift** = grow from **centre** not edge
 - Help appears at the bottom of the screen

Ellipse: 24.13 px × 36.20 px; with **Ctrl** to make square or integer-ratio ellipse; with **Shift** to draw around the starting point

Click or click and drag to start a path; with **Shift** to append to selected path. **Ctrl+click** to create single dots (straight line modes only).

Control nodes

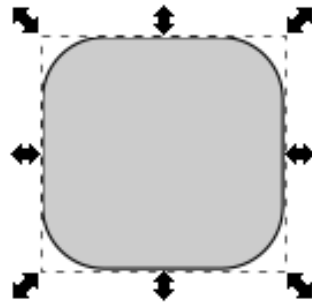
- Use the Edit Nodes tool 
- Two types of control points, squares and circles
 - Squares generally change the size of the shape
 - Circles change the appearance



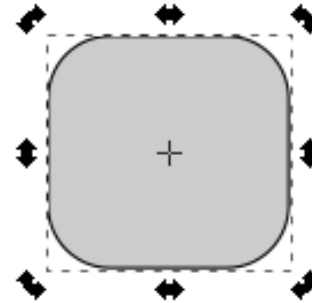
General Transformations

- Select the **Selection** tool
- **Moving** – Click and drag an object
- **Duplicating** – Select object and press Ctrl+D
- **Resizing / Rotating**
 - Click on object
 - Click again to change control arrow type
 - Click and drag arrows to resize / rotate

Resize / Rotate



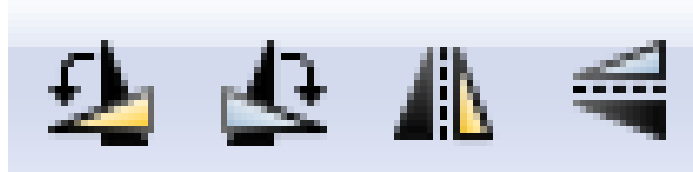
Resize



Rotate

- Can use shift/control keyboard modifiers as before
- For rotation you can move the crosshair to change the centre of rotation

Transform Shortcuts



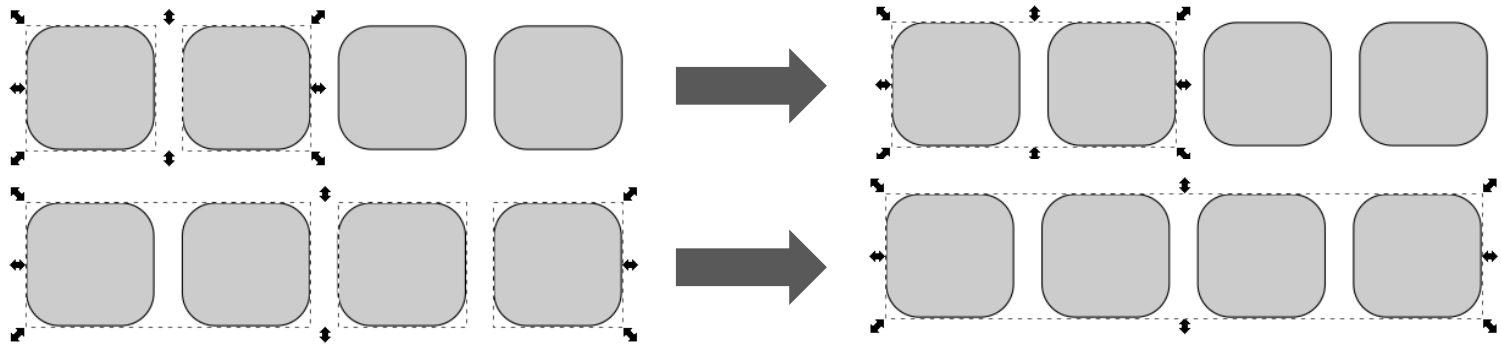
- **Rotate 90 degrees anticlockwise** عكس اتجاه عقارب الساعة
- **Rotate 90 degrees clockwise** مع اتجاه عقارب الساعة
- **Mirror object around the vertical axis** انعكاس رأسي
- **Mirror object around the horizontal axis** انعكاس أفقي

Selecting and Grouping

- Selecting multiple objects
 - Drag a box to cover multiple objects
 - Shift+click to add an object to a selection
 - Shift+click on a selected object to remove it from the selection
- Grouping
 - Combine multiple objects into a single object
 - Reversible

Grouping

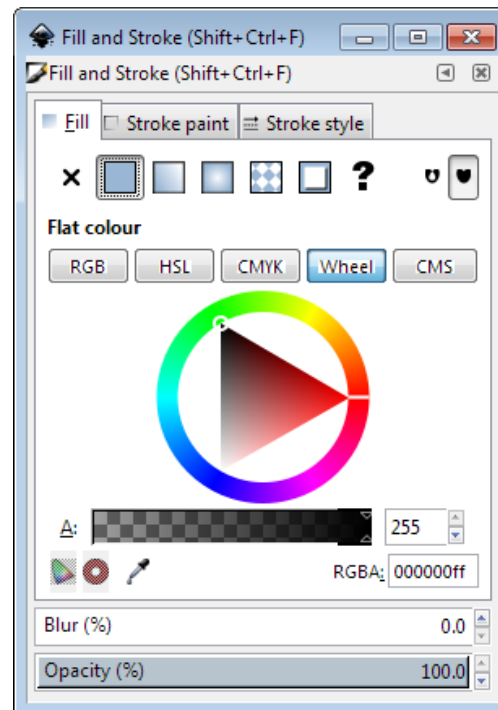
- Select Multiple Objects
- Object > Group (Control+G)
- Multiple Levels



- Object > Ungroup (Control+Shift+G)

Fill and Stroke

- Fill = Colour/Gradient/Pattern within a shape
- Stroke = The line around a shape
- Object > Fill and Stroke (Shift+Ctrl+F)
- Edit
 - Colours
 - Opacity
 - Blur



Fill and Stroke

- Fill types
 - Flat colour
 - Linear gradient
 - Circular gradient
 - Pattern fill
 - Swatch (bitmap) fill
- Specify colour and alpha (opacity)

Flat colour

RGB HSL CMYK Wheel CMS

R: 165
G: 28
B: 28
A: 255

RGB HSL CMYK Wheel CMS

H: 0
S: 181
L: 97
A: 255

RGB HSL CMYK Wheel CMS

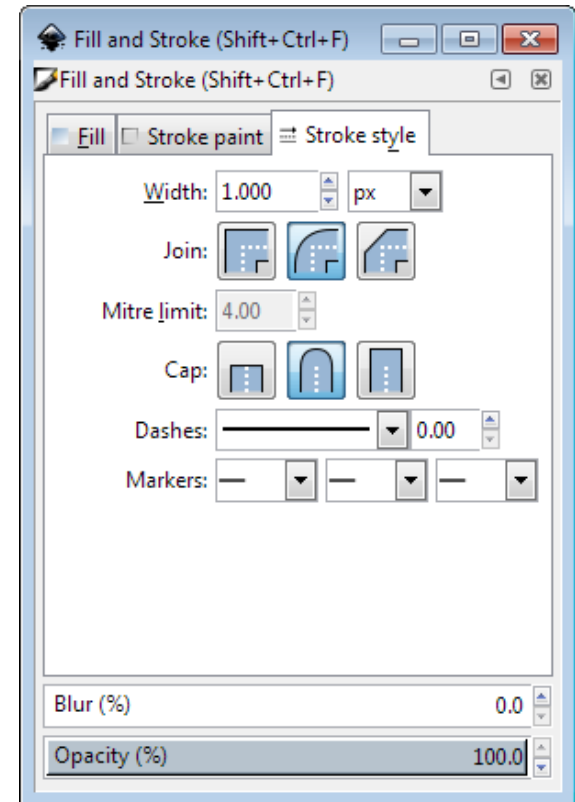
C: 0
M: 83
Y: 83
K: 35
A: 100

RGB HSL CMYK Wheel CMS

A: 255

Stroke Options

- Width of line
- Shape of corners
- Shape of line ends (caps)
- Dashes



Opacity / Blur

- Applies to whole object
- Separate from alpha in colours
- Works on all Objects



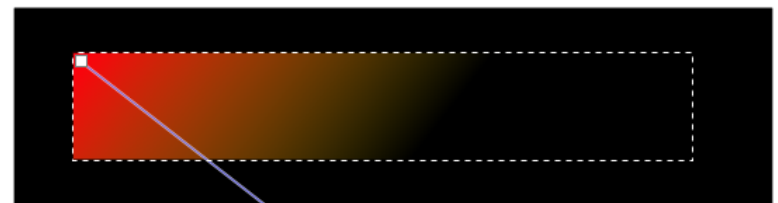
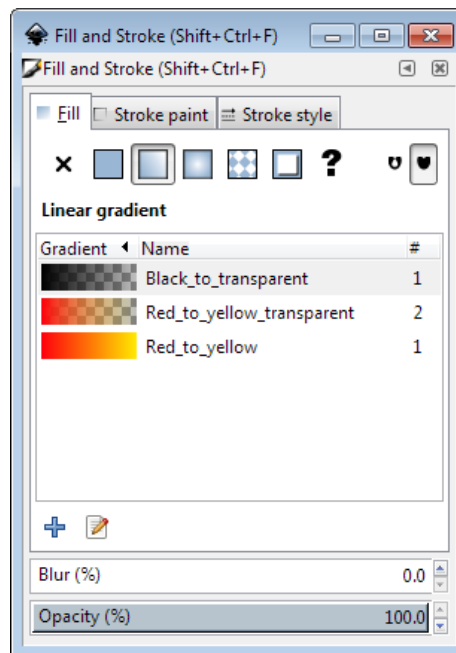
Opacity



Blur
Blur
Blur

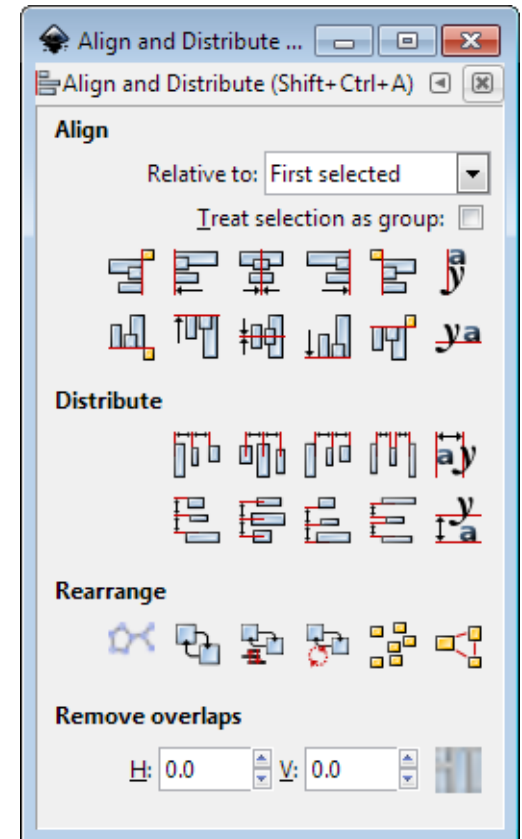
Gradients

- Standard colour option
- Set multiple colours / opacities to go through
- Set the direction and extent of the gradient

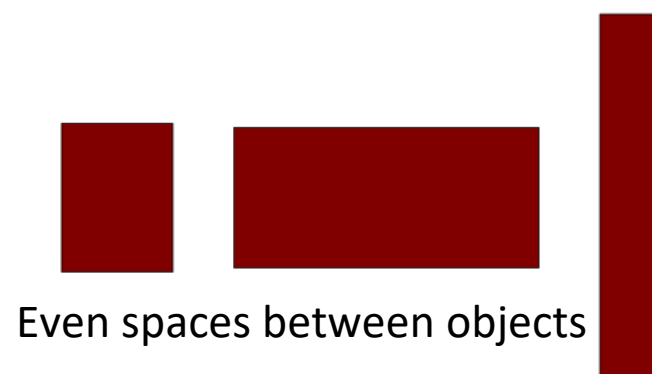
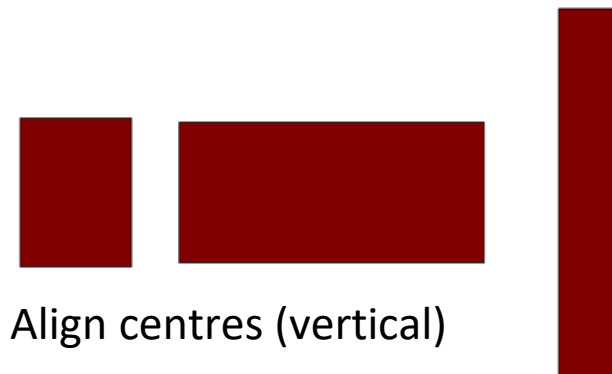
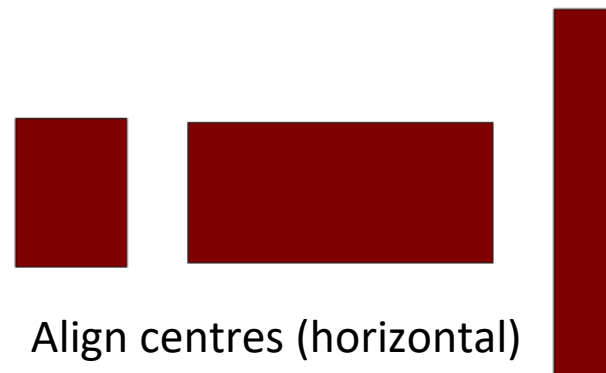
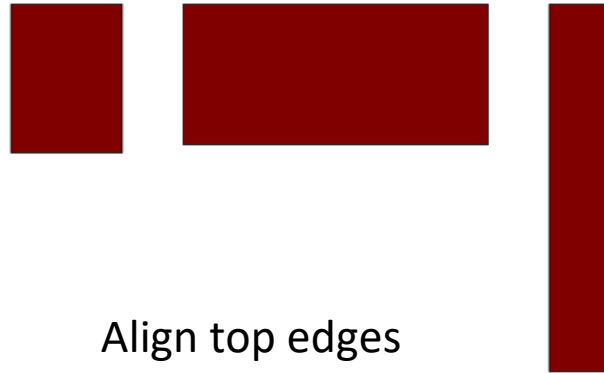


Aligning and Distributing

- Object > Align and Distribute
 - Align = Give objects the same centre/edge position
 - Distribute = Space objects evenly
- Align relative to
 - First/Last selected object in group
 - Largest/Smallest object in group
 - Page
 - Drawing
- Never align anything by eye!

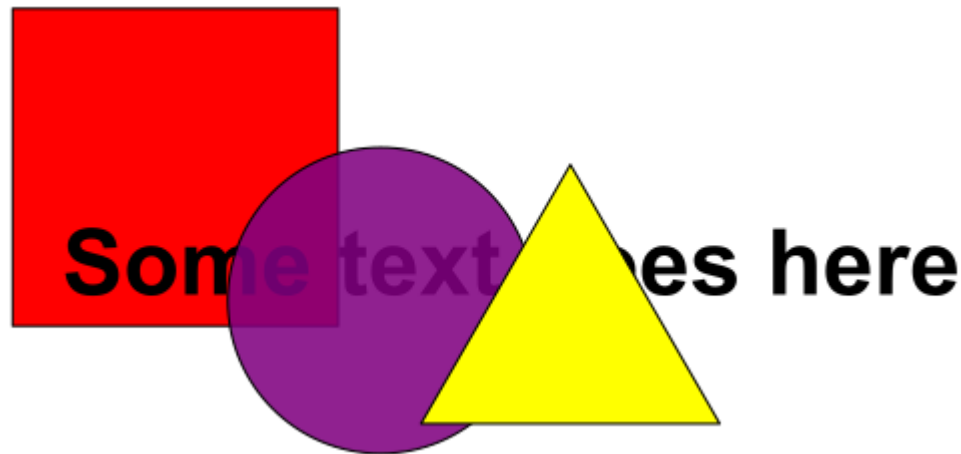


Aligning and Distributing



Z axis - Ordering

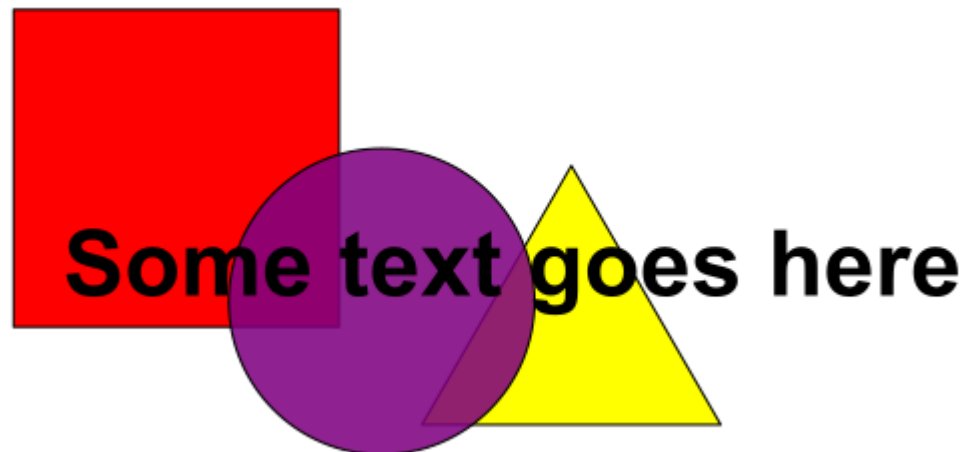
- New objects sit over the top of old objects
- Objects obscure those underneath them (except for transparency)



Z-axis Ordering



- Send object to bottom of z-stack
- Lower object one level
- Raise object one level
- Bring object to top

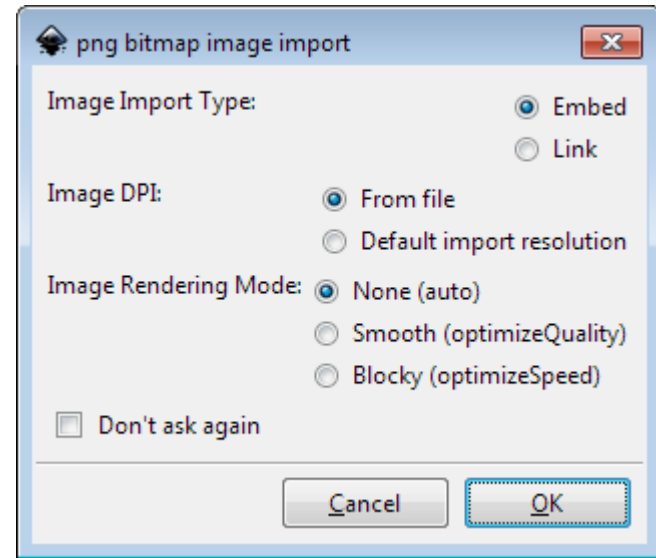


Working with bitmaps (photos)

- Inkscape can include bitmaps in images
- Appear as objects alongside vector objects
- Can't edit the images
- Can't increase the resolution of the image
- Transparency (from PNG etc) is preserved
- File > Import
 - PNG, JPEG, SVG, PDF etc. etc.

Working with bitmaps (photos)

- Embed
 - Large file sizes
 - No updating image
 - Portable file
- Link
 - Small file sizes
 - Need to keep original
 - Can update original
 - Can't (easily) move file



Adding Text

- Use the text tool to add text. 
- Click and type to generate text
- Text can be scaled or rotated as any other object
 - Always hold Ctrl when scaling otherwise the aspect ratio will be messed up

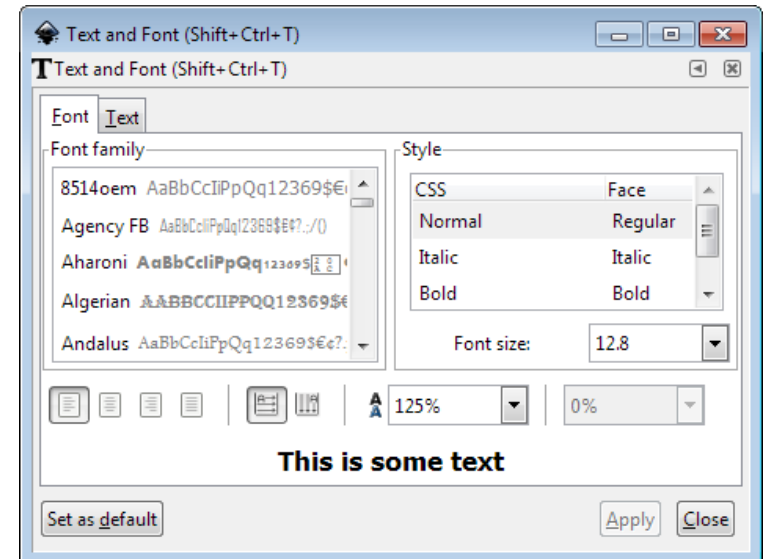
Text options

- Text toolbar



- Text and font settings **T**

- Font
- Alignment
- Spacing

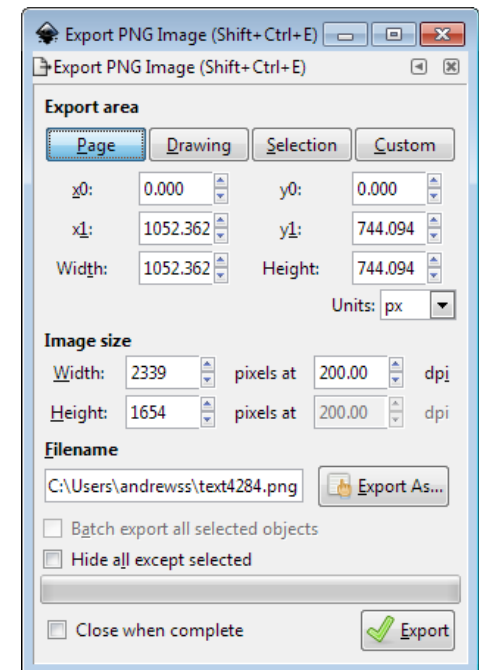
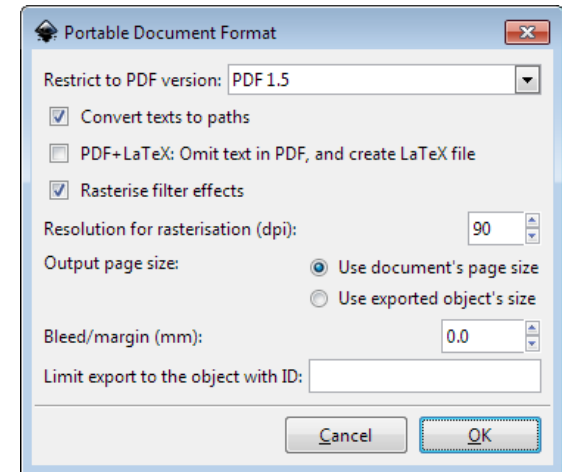


Saving

- Saving
 - File > Save (As)
 - Default is Scalable Vector Graphics (SVG)
 - Adds custom extensions to SVG standard
 - Can select plain SVG for maximum compatibility
 - Can compress SVG (svgz) for disk space saving

Exporting

- Vector
 - Can export as a PDF
 - File > Save (As), select PDF
- Bitmap
 - Can render high quality PNGs
 - File > Export PNG Image



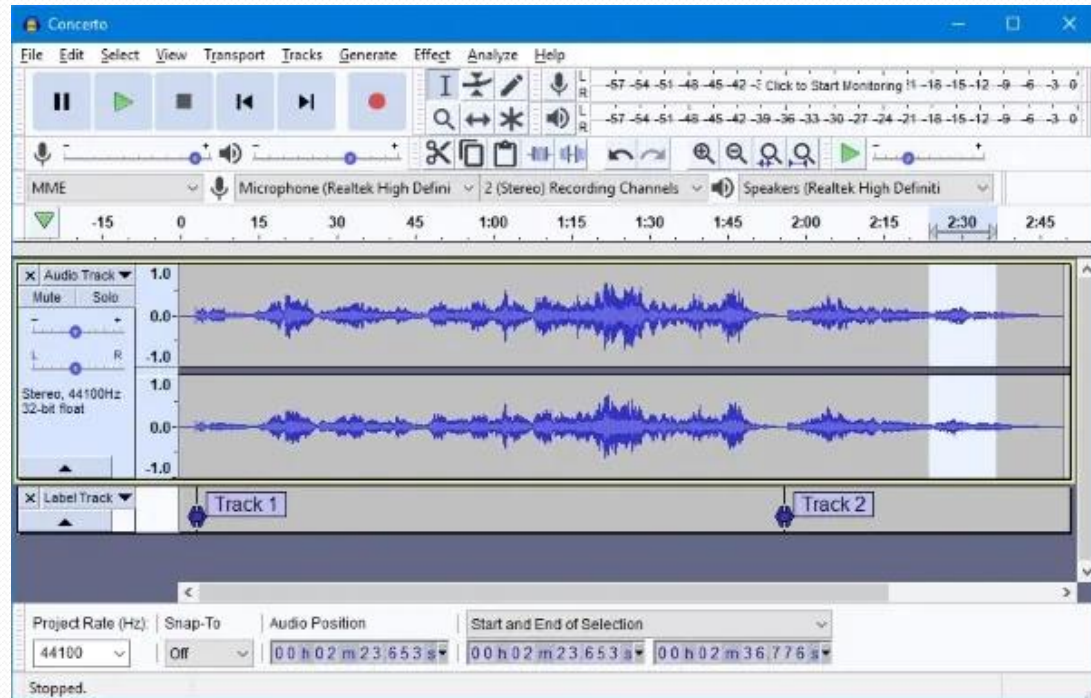
Introduction to Digital Audio

Introduction to Sound

- Sounds are vibrations that travel through the air or some other medium
- A sound wave is an audible vibration that travels through air or water.
- The loudness or intensity of a sound is measured in Decibels (dB).
- The frequency of a sound is measured in Hertz (Hz).
1000 Hertz=Kilohertz (kHz)

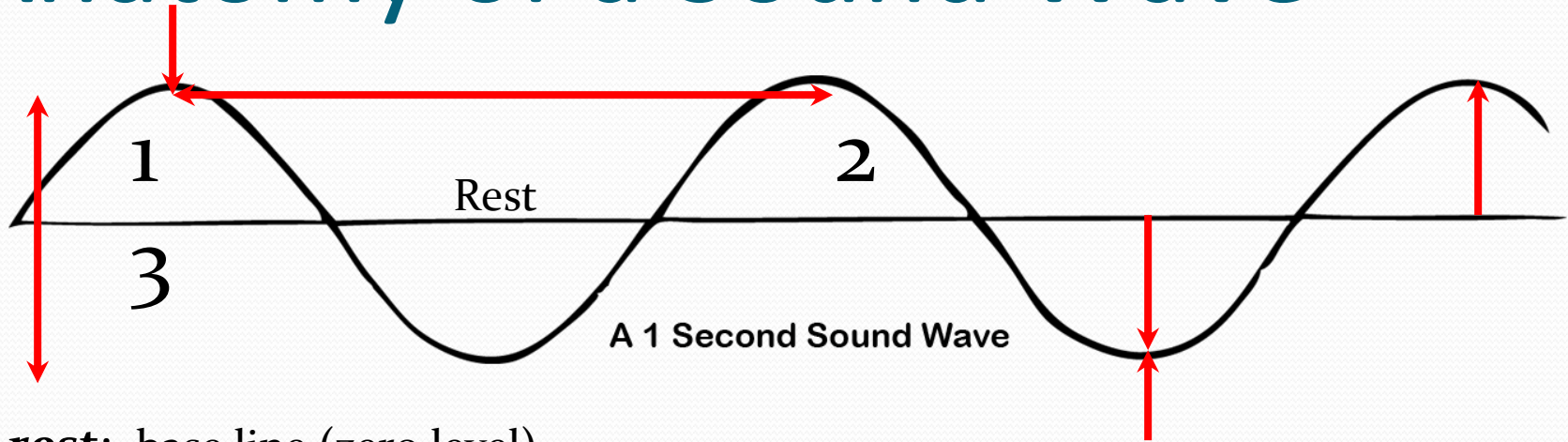
open source editors examples

1. [Audacity](#)
2. [WaveShop](#)
3. [SoundEditor](#)
4. [Ardour](#)
5. [Jokosher](#)
6. [Traverso DAW](#)



<https://filmora.wondershare.com/audio-editing/open-source-audio-editor.html>

Anatomy of a Sound Wave



rest: base line (zero level)

Crest: Point that exhibits **maximum amount of upward** displacement from the rest position.

Trough: Point that exhibits **maximum amount of downward** displacement from the rest position.

Wavelength: Distance from **crest** to **crest**.

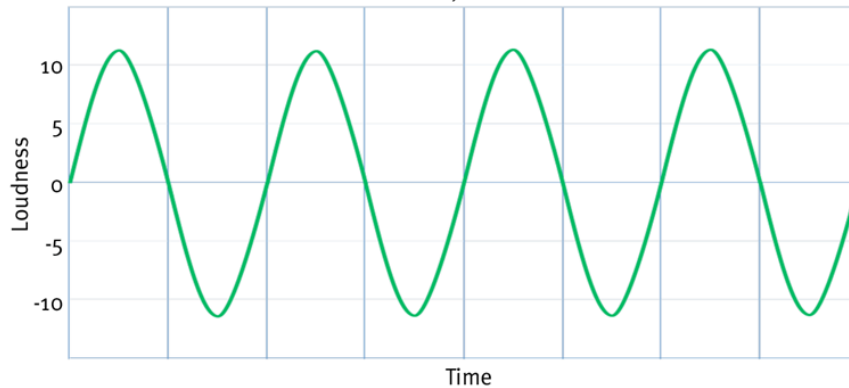
Frequency: Number of waves to pass a point in one second. (How **Fast**)

Amplitude: Distance from **rest** to **crest** or **rest** to **trough**.

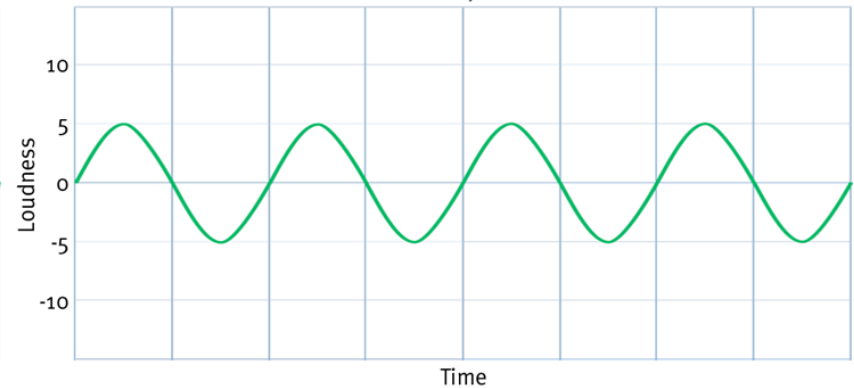
Pitch: Quality of a sound determined by the rate of vibration. The degree of highness or lowness of a tone. (How **High**)

AMPLITUDE

The taller the curves, the louder the sound

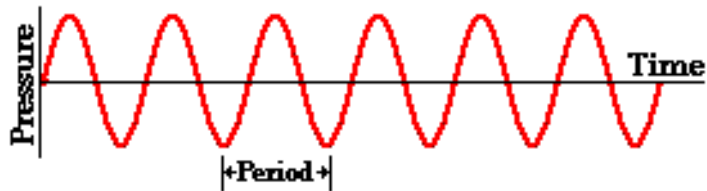


The shorter the curves, the softer the sound

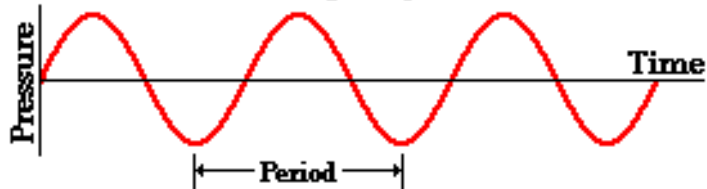


FREQUENCY

High Frequency Wave

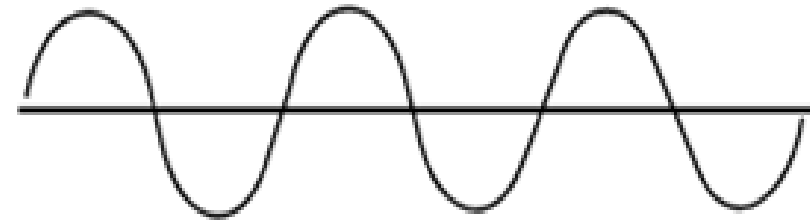


Low Frequency Wave

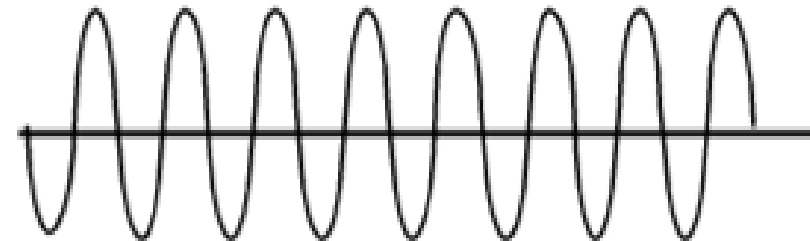


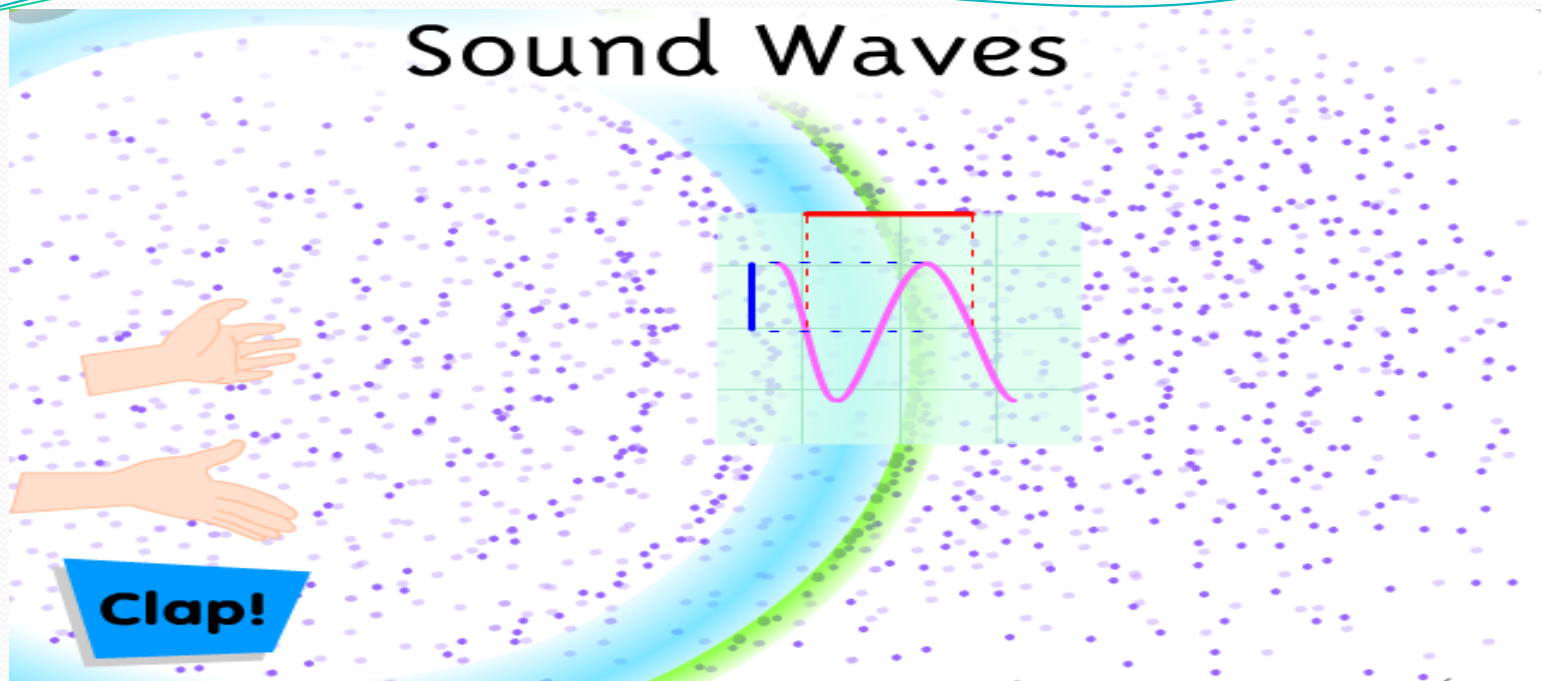
PITCH

Low Pitch
(Waveform of
Bass Guitar)



High Pitch
(Waveform of
Whistle)



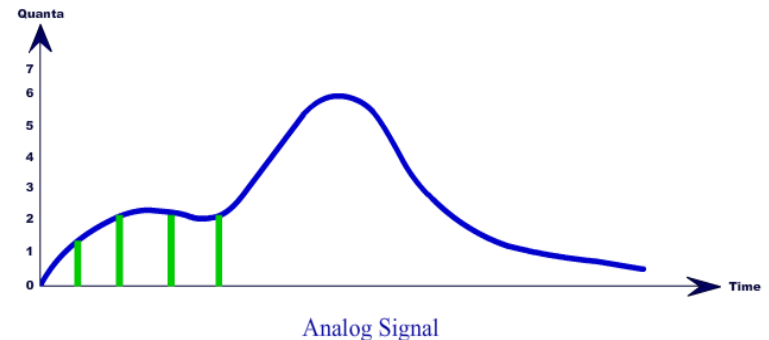


Good video to explain sound properties:

<https://www.khanacademy.org/science/ap-physics-1/ap-mechanical-waves-and-sound/introduction-to-sound-waves-ap/v/sound-properties-amplitude-period-frequency-wavelength>

Converting Analog Sound to Digital Sound

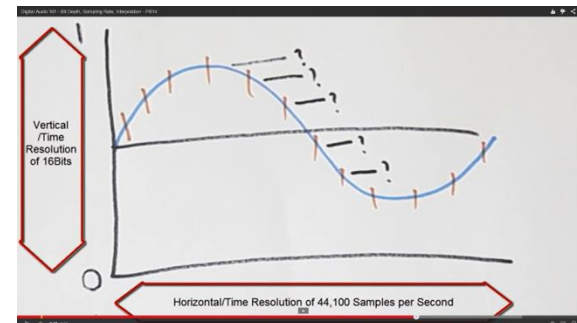
- Analog sound is a continuous phenomena, such as speaking or an electrical pulse from an instrument.
- Computers can only deal with discrete information, so analog sound must be converted to digital, which is called sampling.
- Sampling converts electrical energy to binary code (1's and 0's) that is readable by computers.



[Click to View Sampling Animation](#)

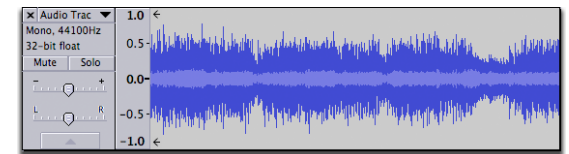
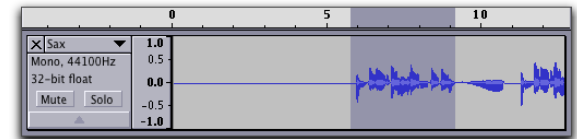
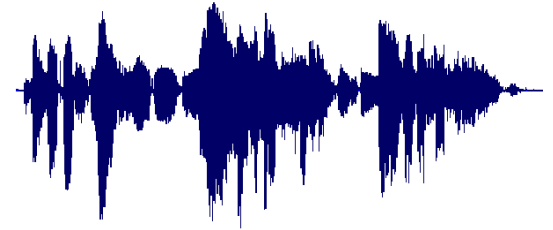
Sampling (Continued)

- Sample Rate: The rate or frequency at which a sound is sampled to digitize it. Measurements of a sound wave over time measured in **Hz (cycles/sec)**.
 - 48 kHz: standard for digital audio
 - CD audio is sampled at 44.1 kHz
- Bit Rate (Depth) (**bits/seconds**): The number of computer bits that are conveyed or processed per unit of time.
 - Normally expressed in kilobits per second (kbps).
 - A higher bit means generally means better quality (resolution).
 - 16 bit: 65,536
 - 24 bit: 16,777,216



Audio Software Terms

- **Waveform:** a visual representation of an audio signal (sound clip).
- **Clip:** is a short segment of audio. It can be combined with others to make an audio track.
- **Track (Channel):** an area upon which audio clips move. It may be a single waveform or could be made up of clips.



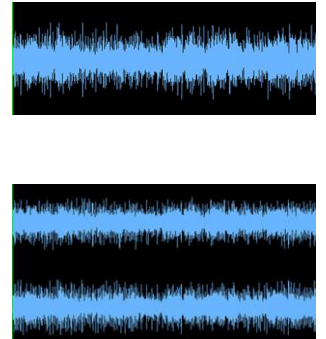
Audio Software Terms (Continued)

- **Channel Types**

- Single Channel=**Mono**
- Two Channels=**Stereo** (use **both ears**)
 - reproduction of sound using two or more independent audio channels in a way that creates the impression of sound heard from **various directions**, as in **natural hearing**.

- **Trim**: removes **non-selected sides** from the **left** and **right** of selected audio.

- **Amplify**: change the volume of selected audio.



Recording Types:

Mono vs Stereo vs 5.1 surround

Sound effects are recorded with various microphone setups. Most of the time they are recorded in:

- **Mono** = recorded with **one microphone**
- **Stereo** = recorded with **two microphones**
- **5.1** = recorded with **six microphones** (.1 is the **bass (main)** channel)

When you are creating your own sounds:

- **DX**, **FOL**, **SFX** should be **mono** files.
- **MX** and **BGs** should be **stereo** files.

Ensure to use **48k sample rate**, **24 bit depth** for **highest quality**

Remember:

- **DX** = Dialogue
- **MX** = Music
- **SFX** = Sound Effects (Hard Effects) - any sound from an real-life object.
- **FOL** = Foley - any sound effect that the player makes.
- **BG** = Backgrounds

2D vs. 3D Audio Sources

For video games, there are two different types of audio sources: 2D and 3D.

1. **2D sounds** = (**non-diegetic** صوت مجاور غير محدد الاتجاه) emanate from player (like wearing headphones)
 - Examples: MX, FOL, most BGs, some DX
1. **3D sounds** = (**diegetic** صوت باتجاه وبعد) emanate from speakers placed around the scene (affected by **distance** and **direction**)
 - Examples: SFX, most DX

AUDIO RECORDING SYSTEMS

When it comes to film and video sound, the first thing you need to determine is whether to use a **single or double** audio recording system.

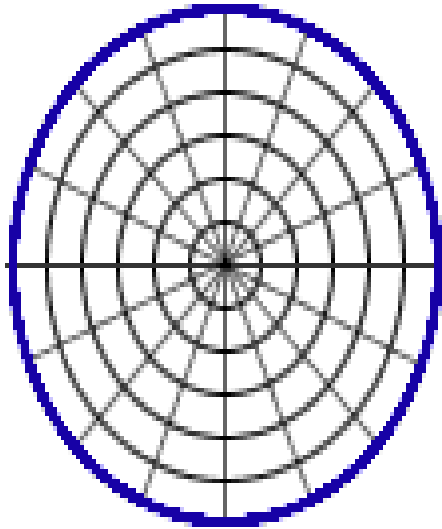
1. In a **single system**, audio is **captured directly into the camera**, and records simultaneously with the image. For a double or dual system, sound is captured independently of the camera and onto a digital audio recorder.
2. **Double system** is ideal if your camera lacks higher-grade audio inputs. But more importantly, a double system delivers stronger audio quality. **this is to user extra sound input device in beside camera.** جهاز اضافي لتسجيل الصوت وتكون الكاميرا للتصوير المرئي فقط.



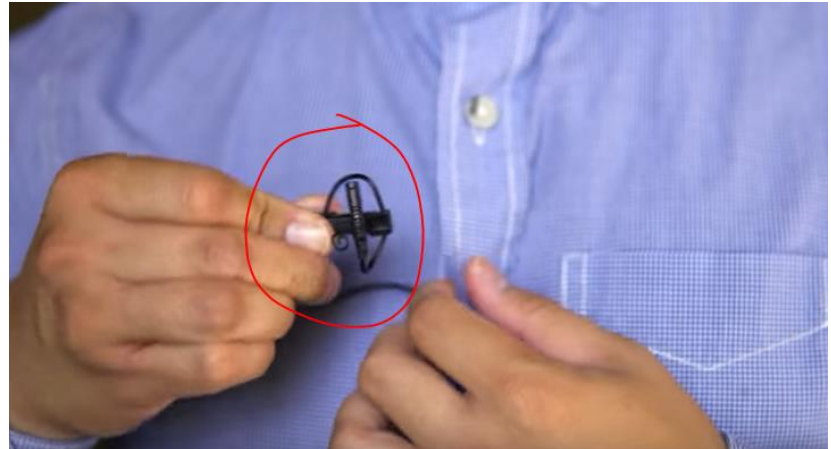
Directional Properties مصدر تسجيل الصوت

- Every **microphone has a property known as *directionality***. This describes the microphone's sensitivity to sound from various directions. Some microphones pick up sound equally from all directions, others pick up sound only from one direction or a particular combination of directions.
- The types of directionality are divided into three main categories:
 - **Omnidirectional**
Picks up sound evenly from **all directions** (omni means "all" or "every").
 - **Unidirectional**
Picks up sound predominantly from **one direction**. This includes cardioid and hypercardioid microphones
 - **Bidirectional**
Picks up sound from **two opposite directions**.

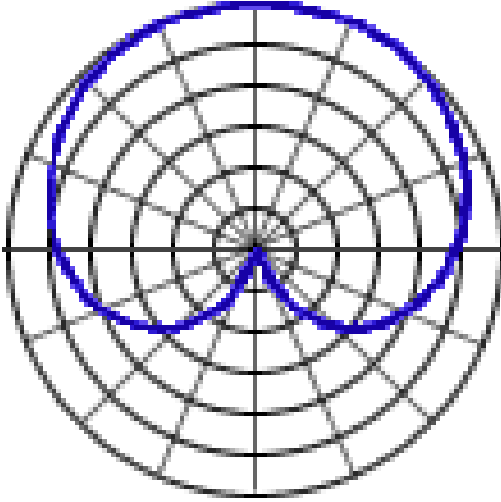
1- Omnidirectional



- **Uses:** Capturing ambient noise; Situations where sound is coming from many directions; Situations where the mic position must remain fixed while the sound source is moving.
- Although omnidirectional mics are very useful in the right situation, picking up sound from every direction is not usually what you need. Omni sound is very general and unfocused - if you are trying to capture sound from a particular subject or area it is likely to be overwhelmed by other noise.
- Example: The **lavalier microphone**, or **lav**, is an omnidirectional microphone that **clips** onto the costume of an actor.

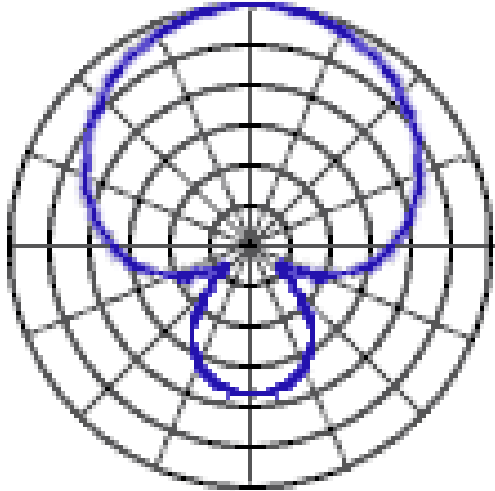


2-Cardioid على شكل قلب a unidirectional mic



- Cardioid means "**heart-shaped**", which is the type of pick-up pattern these mics use. Sound is picked up mostly from the front, but to a lesser extent the sides as well.
- Uses: Emphasizing sound from the direction the mic is pointed whilst leaving some latitude for mic movement and ambient noise. التركيز على مصدر رئيسي للصوت مع وجود اصوات خفيفة محيطة
- The cardioid is a very versatile microphone, ideal for general use. Handheld mics are usually cardioid.
- There are many variations of the cardioid pattern

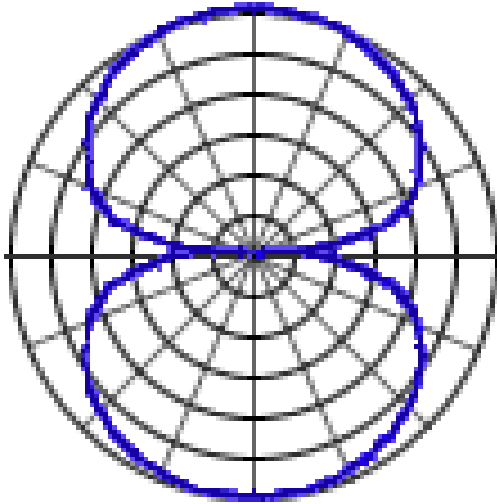
3-Hypercardioid a unidirectional mic



- This is exaggerated version of the cardioid pattern. It is very directional and eliminates most sound from the sides and rear. Due to the long thin design of hypercardioids, they are often referred to as shotgun microphones.
- Uses: Isolating the sound from a subject or direction when there is a lot of ambient noise; Picking up sound from a subject at a distance. تقليل الضجيج العالي مع عدم ازالته كلياً.
- By removing all the ambient noise, unidirectional sound can sometimes be a little unnatural.
- You need to be careful to keep the sound consistent. If the mic doesn't stay pointed at the subject you will lose the audio.
- Shotguns can have an area of increased sensitivity directly to the rear.



4-Bidirectional



- Uses a figure-of-eight patterns (8 Shape figure) and picks up sound equally from two opposite directions.
- Uses: As you can imagine, there **aren't** a lot of situations which require this polar pattern. One possibility would be an **interview with two people facing each other** (with the mic between them).

Exporting Files

Exporting: the process of saving an audio file in a format other than the format of the program it was created in.

Common Audio Formats

● **WAV (.wav)**

- Developed by IBM and Microsoft for Windows
- High Quality (Lossless)
- Large File Size

● **AIFF**

- Developed by Apple
- High Quality (Lossless)
- Large File Size

● **MP3 (.mp3)**

- Compressed format
- Popular for music (MP3 Players)
- Well suited for Internet

● **Ogg Vorbis (.ogg)**

- Open Source
- Compressed format
- Well Suited for Internet
- Popular with Gaming Industry

Comparison between 7 common Audio Tools للاطلاع فقط

	Best for	Standout feature	Pricing
Adobe Audition	Powerful features at a price	Industry-standard audio editing	\$20.99/month (Single app), \$54.99/month (Creative Cloud)
Audacity	Free audio editing	Open source but very powerful	Free
Acoustica Standard Edition	A budget alternative to Adobe Audition	Complex effect chains	\$59.90 (Standard Edition), \$199.90 (Premium Edition)
Hindenburg Pro 2	Journalists and podcasters	Auto-leveling	\$12/month (Standard), \$15/month (Plus), \$30/month (Premium)
SOUND FORGE Audio Studio 17	Windows users on a budget	Video integration	\$59.99 (Single License), \$2.99/month (Audio Studio 365)
Audiate	Editing recordings like a text document	Transcription-based editing	\$29.99/month or \$199.99/year
AudioMass	Browser-based audio editing	Simple interface	Free



AUP

- A file extension for a project file used by **Audacity** audio software.
- *They contain audio project information and settings, and correspond with a **.DAT** file of the same name that contains that audio data referenced in the project file.*
- *AUP files can only be opened by Audacity*
- *To convert an AUP file to another audio format, Select "File," and "Export as MP3" (or another file format).*

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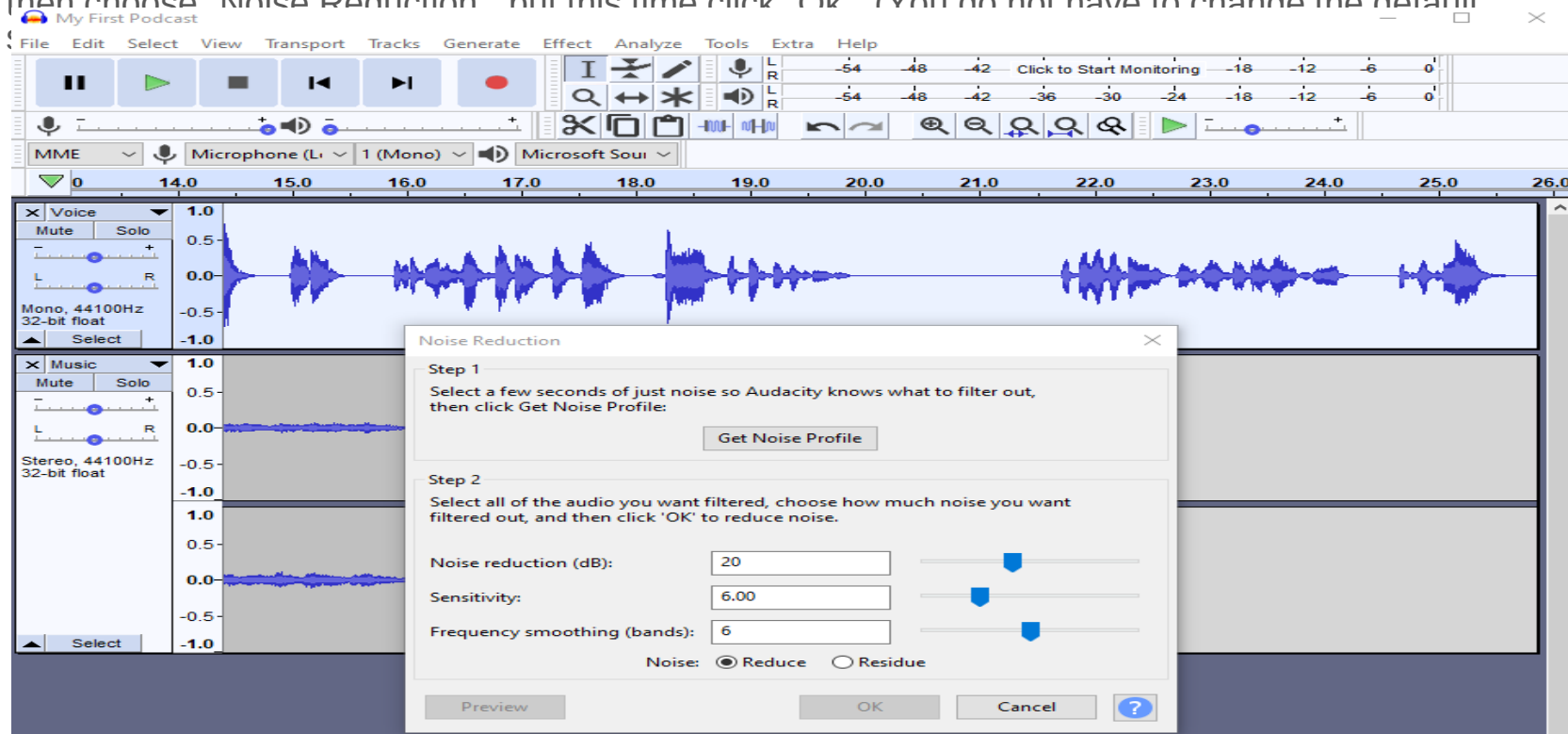
- Open Source
- Compressed format
- Well Suited for Internet
- Popular with Gaming Industry

Audacity Basic Operations

Here are some simple edits you might want to make to your recorded audio in Audacity to quickly improve the audio quality.

1- Remove background noise.

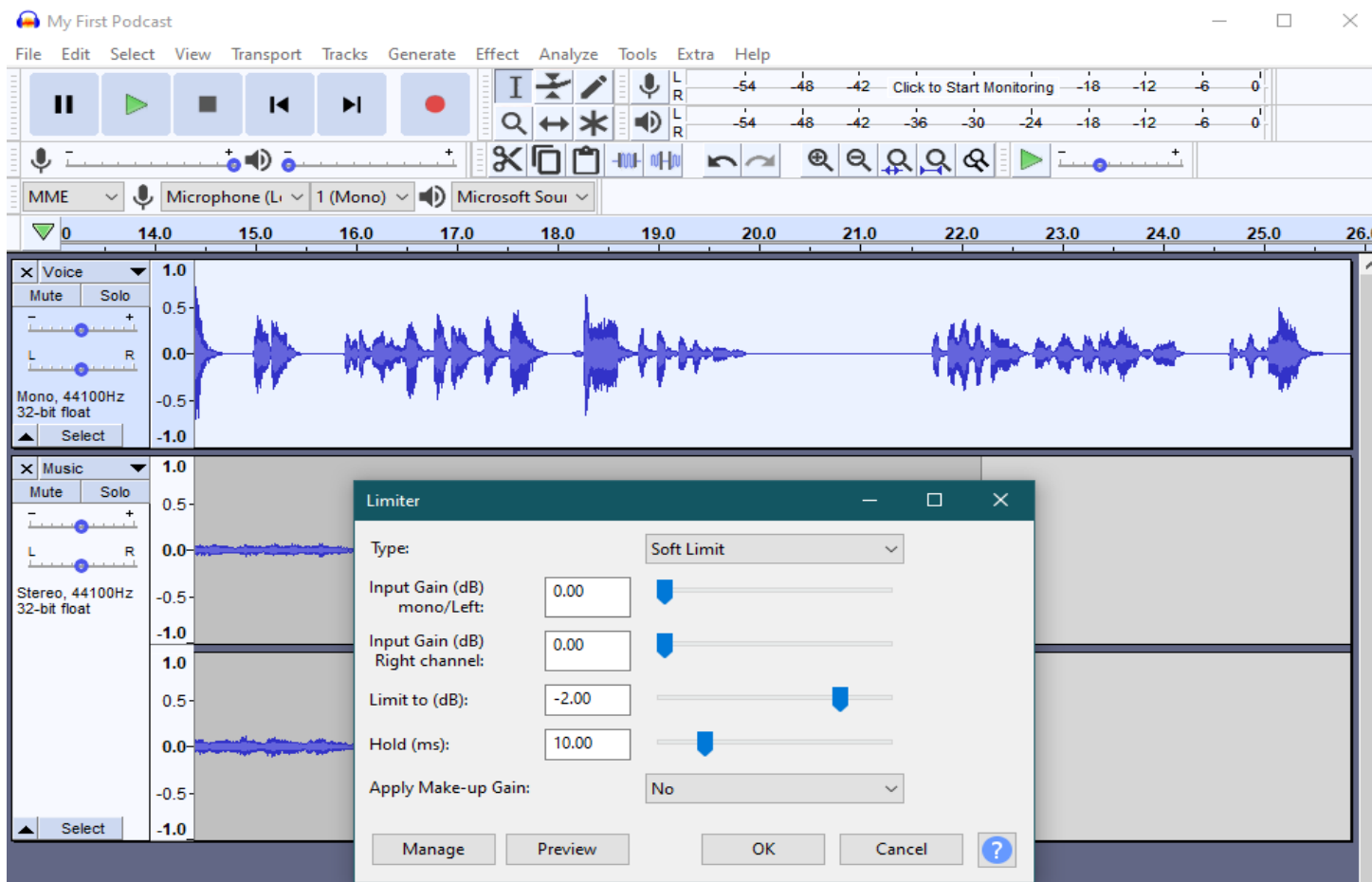
When you record, leave between 5 and 30 seconds of silence in the track. This will pick up the background noise of your recording location, also known as "room tone." It is easy to remove the background noise of the entire track in Audacity once you have a "silent segment." First, highlight the silent segment. Make sure you do not accidentally include any speech. Click on the "Effect" menu, then choose "Noise Reduction." Click on the "Get Noise Profile" button. You will automatically be taken back to the main screen. Next, select the entire track. Once more, click on the "Effect" menu, then choose "Noise Reduction " but this time click "Ok " (You do not have to change the default



Audacity Basic Operations-2

2- Even out the volume.

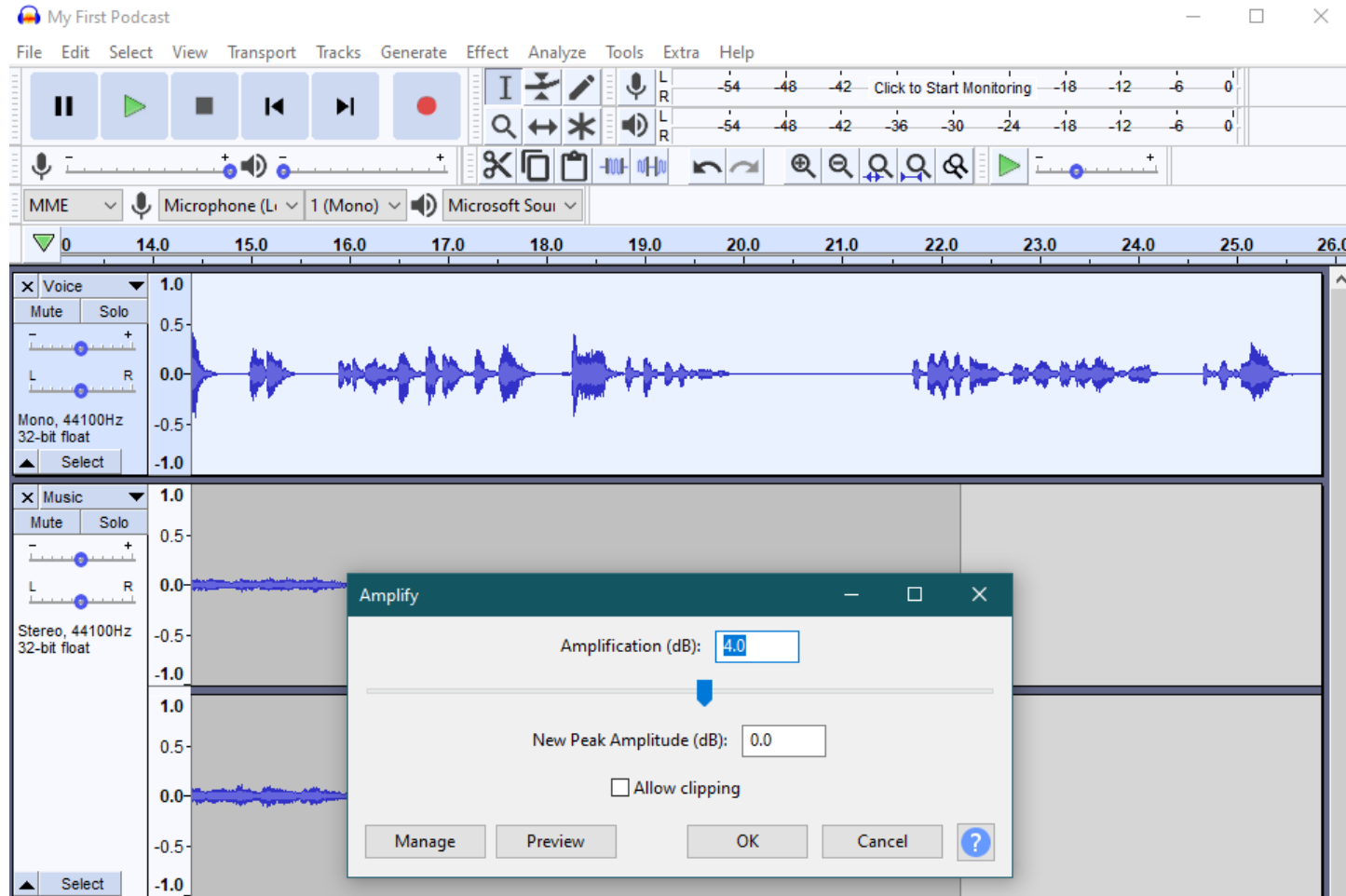
There will be moments in your recording that are much higher than the rest of the audio, such as laughter or exclamations. These moments can be jarring for your listeners. Fortunately, it is easy to fix these sound peaks in Audacity. First, select the entire voice track. Click on the "Effect" menu, then choose "Limiter." You can leave the default settings, but if you want to modify the peak reduction, change the "Limit to (dB)" option. The greater the negative number, the more you flatten the peaks and even out the volume. Be careful not to overdo it, though! Leave some volume variation in the track so it sounds natural. When you have chosen your level, click "Ok."



Audacity Basic Operations-3

3-Change your voice volume.

Sometimes your voice will be too low or too high in the original audio. Although it is better to set the right volume while recording, you can modify the volume of your voice in Audacity easily. First, select the entire voice track. Click on the "Effect" menu, then choose "Amplify." Audacity will make a suggestion in the "Amplification (dB)" box. You can leave this default setting or change it manually. Positive numbers will increase the volume; negative numbers will decrease the volume. We suggest *not* checking the "allow clipping" box, so your audio sounds cleaner. When you have chosen your amplification level, click "Ok."



Where do I find sound effects?

- <http://www.mediacollege.com/downloads/sound-effects/>
- <http://www.flashkit.com/soundfx/>
- <http://www.audiomicro.com/>
- <https://freesound.org/>
- <http://freemusicarchive.org/>
- Record them yourselves with your phone!

Video Editing Introduction

Open Source Tools for video editing

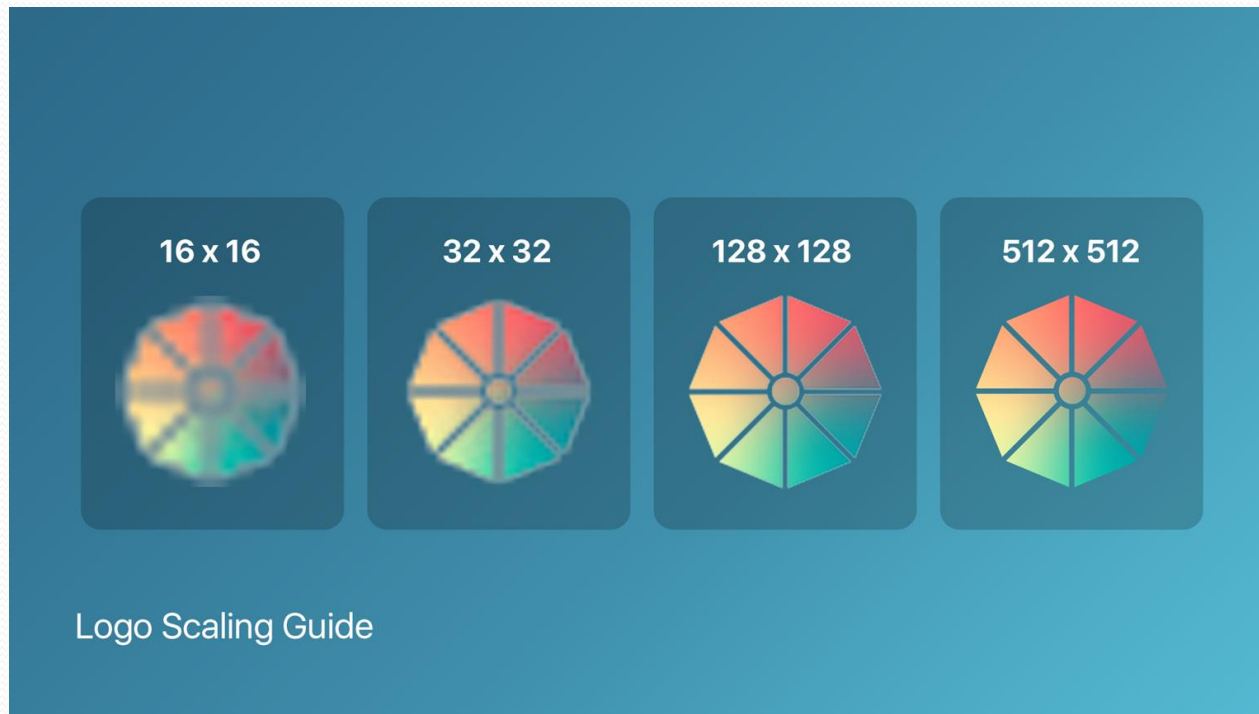
1. **OpenShot**
2. **Shotcut**
3. Kdenlive
4. Flowblade
5. Pitivi
6. **Blender**
7. VLMC
8. **OBS Studio**
9. Olive

Video Resolution

- Resolution has an overall effect on the image quality of videos.
- The higher the resolution, the clearer a video will be.
- Resolution also affects the file size of videos.
- Because of this, a High Definition video has a bigger file size compared to a Standard Definition video with the same duration

The dimensions of video resolution

- Video resolution is measured as **width by height** using pixels as units.
- The following image shows how more pixels make the image sharper and more detailed. To put it simply, more pixels make up better image resolution



Video resolution and corresponding file sizes on Android

Short Name	Name on Device	Video Resolution	File Size (1 minute)
Ultra HD or 4K	3840 x 2160p	3840 x 2160	320 MB
Full HD	1920 x 1080 / 1080p	1920 x 1080	149MB
HD	1280 x 720 / 720p	1280 x 720	105 MB
WVGA / (Wide SD)	720 x 480 / WVGA	720 x 480	26 MB

HD Resolutions



HD Resolutions-2

720 resolution (HD)

This is the **lowest HD resolution** to still be considered HDTV and is often called simply “HD.” Most videos are shot in at least 1080, but 720 (1280 x 720 pixels) can be an acceptable resolution for smaller web content. However, now that most computer screens are HD, best practice is to aim for a higher resolution than 720 for web use and streaming.

1080 resolution (full HD)

Often referred to as “**full HD**,” 1080 (1920 x 1080 pixels) has become the industry standard for a crisp HD digital video that doesn’t break your storage space. This is also a common screen resolution for smartphones.

2K resolution or QHD (**quad high definition**)

The next steps up are QHD (2560 x 1440 pixels) or 2K resolution (2048 x 1080 pixels).

These formats provide more room for image edits, larger displays, and reframing without

lost quality.
https://www.adobe.com/mena_en/creativecloud/video/discover/video-resolution.html

HD Resolutions-3

4K resolution (ultra HD)

Called 4K and marketed often as UHD (ultra-high-definition television), this resolution is technically 3840 x 2160 pixels. It looks quite similar to 2K to most viewers but gives filmmakers more room to zoom in and edit.

8K resolution

Videographers rarely need to shoot in 8K (7680 x 4320 pixels), but this extremely high-res option leaves the most room for creating amazing effects or zooming into a faraway shot without pixelation.

Codec

- It **compresses** and **decompresses** (encodes and decodes) the video – with varying degrees of quality depending on the codec. There are two types of codecs: lossy and lossless.
- **Lossy codecs** compress the video by losing details of the video. If a video is heavily compressed, the result is poor picture quality.
- **Lossless codecs**, on the other hand, compress the video without losing details – so the quality of the original is fully preserved. However, lossless codecs are usually very resource-intensive and the video becomes very large as a result.

<https://cleverclipstudios.com/en-ch/blog/videoformats/>

Some Known Codecs للاطلاع فقط

Codec	Description	Memory consumption	Resolution	Speed
H264 (GPU)	Universal codec, used for most video clips	34 MB	Medium	High
HEVC (GPU)	Popular for high compression, tasks up less memory than H264	33.5 MB	Medium	High
Xvid	Favorite among many creators due to its high quality	55 MB	High	Medium
H264	High-quality codec, although larger than most	217 MB	Very high	High
MPEG-1	While the codec doesn't offer knife-sharp quality, it leads by compression rates.	38.8 MB	Medium	Below average
RGB 24	Low-size, lossless compression	104 MB	High	Very high

<https://blog.pics.io/which-video-format-is-better-breakdown-of-common-video-file-types-6737ae635081/>

Video Formats-1

1. MP4

MP4 is the golden mean among the types of video files when it comes to file size, loading speed, and compression.

Suitable for **streaming**, supports overlay texts, and ensures **lossless compression**. That's why it became widely popular on the **web**. Now, most **websites, social media** platforms, and multimedia players use it as an almost universal standard.

Pros:

- High degree of **compression**,
- Contains metadata,
- Versatile, with two extensions (mp4a and mp4v).

Cons:

- Codec-depending,
- MP4 file formats don't protect creators from copyright infringement.

Video Formats-2

2-MKV

MKV can store **captions** تعليقات كتابية, **audio**, and **video as a single file** making it a popular option for playback of lengthy videos such as movies. Due to the **larger file size in comparison** to MP4, however, MKV is more frequently used for **downloads and DVD/Blu-Ray playback** rather than streaming.

Pros:

- **Open-source** offers more editing freedom,
- High video quality,
- Supports **all codecs**.

Cons:

- **Few mobile** players support MKV,
- **Slow loading** speed,
- **Large** video files.

Video Formats-3

AVI

5–7 years ago AVI used to be one of the most popular video formats out there. Although it got dethroned over time, the video type still gets credit for its high compatibility. For instance, it's one of the few formats you can run **both on Windows, Mac, and Linux** devices.

The main issue with AVI is its **poor compression** as it severely damages the quality of the video. So, AVI files are usually larger than the rest making it a better option for storage, not so much for streaming.

Pros:

- Keeps the audio **quality high** even after compressing the clip,
- There's a wide range of AVI-**supporting codecs**,
- One of the top DVD management formats.

Cons:

- File compression results in visible **quality losses**,
- Unreliable soft **subtitle** support.

Video Formats-4

MOV

MOV is known as the “video editing software **friendly format**” among different file types. It was designed by Apple .

Pros:

- **Compatible** with most formats.
- Facilitates HD video making.

Cons:

- **Large file size.**
- **Lossy video** compression.
- Not compatible with all popular **players** (programs to play videos).

Video Formats-5

WEBM

WEBM is a **Google-backed video format** created for web videos. The format supports HTML5 and focuses on keeping clips as small as possible — this way, it won't slow the page down.

Pros:

- **Open-source** and free of royalty licenses.
- **Lossless** compression .
- **Small** file size.

Cons:

- **Less** quality than MP4 .
- Limited support for **mobile devices**.
- **No Mozilla Firefox** support.

<https://blog.pics.io/which-video-format-is-better-breakdown-of-common-video-file-types-6737ae635081/>
<https://multimedia.easeus.com/video-converter-tips/webm-vs-mp4.html>

Which Video Format is Right For You?

	MP4	MKV	AVI	FLV	MOV	AVCHD	WEBM
Pros	High degree of compression Contains meta-data Versatile, with two extensions (mp4a and mp4v).	Open-source offers more editing freedom High video quality Supports all codecs.	Keeps the audio quality high even after compressing the clip. Wide range of supported codecs. One of the top DVD management formats.	Used for all types of interactive content. Common among website managers. Used as a backbone for basic games.	Simple container design. Compatible with most formats. Facilitates HD video making.	Impressive image quality. Leading format for blu-ray videos. First-rate sound quality when combined with the Dolby Digital AC-3 codec.	Open-source and free of royalty licenses. Lossless compression Small file size.
Cons	Codec-dependency Relatively high loading time Doesn't protect creators from copyright infringement	Few mobile players support MKV Slow loading speed Large video files	File compression results in visible quality losses. Unreliable soft subtitle support No user interactivity support	Decreasing popularity. No iOS support. High video making cost.	Large file size. Lossy video compression Not compatible with all popular players.	Large files, the size can stretch up to 1000MB. Not compatible with most players, as well as mobile devices.	Relatively low video quality. Limited support for mobile devices. No Mozilla Firefox support.
Applications	Social media, web	DVD, web	DVD	Web, interactive content	Social media	Blu-ray	Web

<https://blog.pics.io/which-video-format-is-better-breakdown-of-common-video-file-types-6737ae635081/>

Social networks and video formats

للاطلاع فقط



Facebook



Instagram



Youtube



Snapchat



LinkedIn



Xing



Twitter

Supported
formats

Platform
founders rec-
ommend MOV
and MP4 but
most common
platforms are
supported

All common
formats are
supported,
MOV and MP4
are recom-
mended

All popular
file types are
Youtube video
formats

MOV and MP4

All top formats
are supported,
the platform
recommends
MP4 and MOV

No specific
restrictions

MOV and MP4

Video
length
limits

240 minutes

60 seconds

12 hours

10 seconds

10 minutes

No limits

140 seconds

<https://blog.pics.io/which-video-format-is-better-breakdown-of-common-video-file-types-6737ae635081/>